

👉 Ensemble Learning — Short Notes

What is Ensemble Learning?

Ensemble Learning combines multiple machine learning models to produce a **stronger and more reliable model**.

Idea: Wisdom of the Crowd → Many models together can perform better than one model.

How does it work?

Multiple **base models** make predictions, and their outputs are combined:

- **Classification** → Majority voting
- **Regression** → Averaging

Important: The base models should ideally be **diverse** (different algorithms or different training data).

Types of Ensemble Learning

Technique	How it works
🗳️ Voting	Different models → combine their predictions
🧠 Stacking	Base models → meta-model learns how to combine them
🛒 Bagging	Same algorithm → different random samples of data
🚀 Boosting	Models are trained sequentially to correct previous errors

1. Voting

Example:

SVM + Logistic Regression + Decision Tree

If predictions are:

Cat, Cat, Dog → Cat wins

Each model generally gets equal importance in basic voting.

2. Stacking

- Multiple base models make predictions.
- A **meta-model** learns how to combine those predictions.
- Unlike simple voting, the meta-model learns which predictions are more useful.

3. Bagging

Bagging = Bootstrap Aggregating

- Create different random samples from the training data **with replacement**.
- Train the same type of model on each sample.

- Combine their predictions.

🌲 **Random Forest = famous example of Bagging.**

4. Boosting

Models are trained **one after another**.

- Model 1 makes predictions.
- Model 2 focuses more on errors of Model 1.
- Model 3 focuses on remaining errors.
- Continue → combine models into a strong learner.

Examples: **AdaBoost, Gradient Boosting, XGBoost**

Why Ensemble Learning works?

- ✅ Improves accuracy
- ✅ Reduces variance
- ✅ Can reduce bias
- ✅ More robust to noise
- ❌ More computationally expensive
- ❌ More complex than a single model

★ Easy Memory Trick

Voting → Different models

Stacking → Meta-model

Bagging → Parallel + different samples

Boosting → Sequential + correct errors